Deterministic–Nondeterministic Interfaces

A Textbook-Ordered Reader of Core Results, Zip–Inverse-Zip Structures, and Mathematical Audit

Collected research materials • 20 October 2026

Editorial scope. This volume organizes three existing documents in a pedagogical sequence: the provisional ATLAS-repaired DNI v0.2 research note; the formal Zip–Inverse-Zip Interface Theory exposition; and the ATLAS DNI reaudit report. The source documents are reproduced as distinct parts, preserving their own statements and qualification of results.

Status. This is a collected reader, not a new independently verified monograph. The central finite-state DNI claims are presented as results of the provisional research note; source-recovery and interface-composition claims in the zip exposition are elementary under their stipulated definitions. General cross-theory compatibility and the OP-01 expedition remain open research directions.

Reading path

Part I — Dynamics and exact interfaces. Begin with deterministic maps, possibilistic successor sets, and stochastic kernels. Learn why exact interfaces require readout and preparation maps satisfying equations, not merely a resemblance between trajectories.

Part II — Provenance-preserving zip and inverse zip. Having learned the meaning of an exact interface, study how separate theories can be assembled as tagged records, recovered without loss, and connected only by separately certified proof-transport maps.

Part III — Independent audit and limits. Finally, read the ATLAS reaudit as an evidentiary companion: distinguish established calculations, acknowledged antecedents, restricted hypotheses, and open extensions.

Guide to the central distinctions

Possibility is not probability. A support relation records which transitions can occur; it discards transition weights. A support-preserving map need not reproduce stochastic laws.

Realization is not ontology. A deterministic hidden-state representation with a stochastic preparation does not establish that a physical system is deterministic.

Provenance recovery is not theorem transport. A tagged zip admits canonical source recovery. Transport of derivability additionally requires a typed translation, proof transformation, assumption

control, and destination verification.

Reachability is not discovery. A target can be reachable by a legal finite path even if a particular deterministic search policy fails to find it within budget.

Structural connectivity is not OP-01 settlement. A graph of source families can be connected while lacking a verified target proof, genuine contributions from the required original islands, or a favorable fully charged cost comparison.

Part I — Exact deterministic, possibilistic, and stochastic interfaces

Primary source: *Deterministic–Nondeterministic Interfaces: Exact Realization, Finite-Resource Obstructions, and Common-Theory Compatibility*, ATLAS-repaired working note v0.2 (8 pages).

Study questions

1. What information is preserved by an exact deterministic–stochastic intertwining, and what additional requirement is needed to preserve the law of complete observable histories?

2. Why can recurrent branching obstruct finite hidden-state deterministic realization, even though unrestricted path-space constructions are available?

3. What does minimum hidden-state cardinality measure, and what does it not measure about computational time, physical mechanisms, or implementation precision?

4. Which results concern a restricted finite classical model, rather than arbitrary independently specified mathematical theories?

Part II — Zip–Inverse-Zip Interface Theory

Companion exposition (4 pages). Read the definitions and proofs as a formal bridge between the Multiarticle record discipline and candidate DNI-style theory interfaces.

Study questions

1. Why is the inverse-zip projection naturally defined on the image of the corresponding source inclusion rather than on all records from unrelated islands?

2. Which proof obligations convert a structural zip edge into a certified theorem-transport interface?

3. When does composition of two interfaces preserve proof judgments, and what happens when their hypotheses or certificate formats are incompatible?

4. Why does an interface network not automatically prove that 26 original source families materially contribute to a settlement?

Part III — ATLAS Reaudit

Review the 3-page reaudit after studying the primary results. Treat it as an assessment of the source note, not as an independent machine-checker receipt unless one is expressly supplied.

Editorial note

The reader is intentionally organized from mathematical definitions and results, to cross-theory application, to critical audit. Page numbering within each reproduced source remains original. The following source documents are preserved rather than silently rewritten or merged at theorem level.

Deterministic–Nondeterministic Interfaces

Exact Realization, Finite-Resource Obstructions, and Common-Theory Compatibility

ATLAS-repaired mathematical working note • Version 0.2 • 20 October 2026

Abstract

We investigate a deliberately restricted interface between deterministic, possibilistic, and stochastic discrete-time theories. A readout/preparation intertwining for deterministic and Markov dynamics is explicitly attributed to classical Markov-function theory. We supply a parallel exact interface for set-valued nondeterminism and show that taking finite stochastic supports respects composition and interfaces. Our central finite-state result characterizes the existence and *minimum hidden-state cardinality* of an exact deterministic realization of a given finite-state Markov chain: such a finite realization exists precisely when every vertex lying on a directed cycle of the positive-transition graph has exactly one possible successor. When this condition holds, the minimum is the number of legal infinite observable paths, computable by a finite recurrence; when it fails, no finite hidden-state realization exists. We give a fully explicit four-state optimal interface, a linear-feasibility test for a fixed implementation, and a total-variation error bound. We distinguish these results from the still-unresolved task of building a general semantic interface between independently specified theories. The contribution is an auditable classical synthesis; no general originality claim or physical interpretation is presumed.

Scope and status. We prove precise results in finite classical transition systems, give standard Borel path-space realizations as background, and formulate rather than solve the unrestricted theory-of-theories problem. The shared formal language is somewhat more general than either deterministic or probabilistic transition systems, but is *not* asserted to subsume all possible operational or physical theories.

1 Three dynamics and the restricted target

For a nonempty finite set Y, let $\mathcal{D}(Y)$ denote probability distributions and $\mathcal{P}_+(Y)$ its nonempty subsets. A deterministic dynamics is $f : Y \to Y$; a possibilistic dynamics is $N : Y \to \mathcal{P}_+(Y)$; and a probabilistic dynamics is a row-stochastic kernel $K : Y \to \mathcal{D}(Y)$. Every f induces both $N_f(y) = \{f(y)\}$ and the Dirac kernel $K_f(y, z) = \mathbf{1}_{\{z=f(y)\}}$. Conversely, finite K induces the nonempty successor relation

$$N_K(y) = \{z \in Y : K(y, z) > 0\}. \tag{1}$$

A possibility relation does not determine weights; this distinction will remain explicit.

Given K and a finite hidden-state budget m, our concrete question is not merely whether one can encode future stochastic outcomes in an unrestricted hidden state. It is whether there exist a finite set X of size at most m, a deterministic $f : X \to X$, readout $R : X \to Y$, and a stochastic preparation $E : Y \to \mathcal{D}(X)$ satisfying exact interface equations. We then ask for the smallest such m, with no bound placed on the numerical precision of E. We do not constrain a prescribed physical f, which would be a strictly harder problem.

2 Exact stochastic interfaces and acknowledged antecedents

For measurable X, Y, let $f : X \to X$ and $R : X \to Y$ be measurable functions, K a Markov kernel on Y, and E a Markov kernel $Y \rightsquigarrow X$. For measures μ, write $g_*\mu$ for pushforward along g; the composite

kernel EK means

$$(EK)(y, A) = \int_Y E(z, A)K(y, dz).$$

Definition 2.1 (Exact deterministic–stochastic interface)**.** The quadruple (f, K, R, E) is an *exact stochastic DNI interface* if

$$R_*E(y) = \delta_y \qquad (y \in Y), \tag{S1}$$
$$f_*E(y) = (EK)(y) \qquad (y \in Y). \tag{S2}$$

Equations are kernel equalities and cannot be weakened to matching only individual outcomes.

In row-matrix notation, letting $P(x, x') = \mathbf{1}_{\{f(x)=x'\}}$, $\Phi(x, y) = \mathbf{1}_{\{R(x)=y\}}$, and $\Lambda = E$, these are $\Lambda\Phi = I$ and $\Lambda P = K\Lambda$. They are the deterministic-upstairs special case of the classical intertwining/projection conditions associated with Rogers–Pitman [1, 2]. Thus the interface identities and Markov-function assertion are *not claimed as new.*

$$\begin{array}{ccc} \mathcal{D}(Y) & \xrightarrow{K_*} & \mathcal{D}(Y) \\ {\scriptstyle E_*}\downarrow & & \downarrow{\scriptstyle E_*} \\ \mathcal{D}(X) & \xrightarrow[f_*]{} & \mathcal{D}(X) \end{array}$$

Theorem 2.2 (Intertwining and full observable history)**.** *If* (S1)–(S2) *hold and Y is finite, then for every $n \geq 0$,*

$$R_*f_*^nE = K^n. \tag{2}$$

Moreover, if $X_0 \sim E(y)$, $X_t = f^t(X_0)$, and $Y_t = R(X_t)$, then

$$\Pr(Y_0 = y_0, \ldots, Y_n = y_n) = \mathbf{1}_{\{y_0=y\}} \prod_{t=0}^{n-1} K(y_t, y_{t+1}).$$

Proof. Induction on n using (S2) gives $f_*^nE = EK^n$; apply R_* and (S1). For joint histories, initially the hidden-state distribution conditional on $Y_0 = y$ is $E(y)$. If the hidden conditional distribution after a positive-probability observed history ending at z is $E(z)$, then after one deterministic step it is $\sum_w K(z, w)E(w)$. By (S1), $E(w)$ is concentrated on $R^{-1}(w)$, so conditioning on the next readout w recovers $E(w)$ with probability $K(z, w)$. Induction gives the complete finite-dimensional law. □

Remark 2.3. Preparing a randomly chosen hidden state and thereafter evolving deterministically does not establish a deterministic physical ontology. The statistical preparation is an explicit assumption, not a philosophical conclusion.

3 A parallel possibilistic interface

For $N : Y \to \mathcal{P}_+(Y)$, let $L : Y \to \mathcal{P}_+(X)$ be a nonempty *set-valued preparation.* For a subset $A \subseteq X$, define $f[A] = \{f(x) : x \in A\}$ and $R[A] = \{R(x) : x \in A\}$.

Definition 3.1 (Exact deterministic–possibilistic interface)**.** The quadruple (f, N, R, L) is an exact possibilistic DNI interface if for all $y \in Y$,

$$R[L(y)] = \{y\}, \tag{P1}$$
$$f[L(y)] = \bigcup_{z \in N(y)} L(z). \tag{P2}$$

These are equalities of sets, not probabilities. Different preparation sets for distinct y are disjoint by (P1).

Proposition 3.2 (Finite support bridge)**.** *If X, Y are finite and (f, K, R, E) is an exact stochastic interface, then (f, N_K, R, L_E) is an exact possibilistic interface, where $L_E(y) = \{x : E(y, x) > 0\}$.*

Proof. Equation (S1) implies $L_E(y) \subseteq R^{-1}(y)$ and nonemptiness. Taking pointwise positive supports of (S2) yields $\mathrm{Supp}(f_* E(y)) = f[L_E(y)]$ and $\mathrm{Supp}((EK)(y)) = \bigcup_{z:K(y,z)>0} L_E(z)$, since all summands are nonnegative. These are (P1)–(P2). □

Proposition 3.3 (Compositional support)**.** *For finite stochastic kernels K_1, K_2, their positive-support relations satisfy $N_{K_1K_2} = N_{K_1} \circ N_{K_2}$ in the corresponding order of relational composition. Dirac kernels map to singleton relations.*

Proof. A matrix product entry $(K_1K_2)(y, z) = \sum_w K_1(y, w)K_2(w, z)$ is positive if and only if some summand is positive. □

Theorem 3.4 (Possibilistic path-space representation)**.** *Every finite, everywhere-nonempty possibility relation N admits an exact possibilistic interface with deterministic left shift on the set $\mathsf{Path}_\infty(N)$ of all legal infinite paths.*

Proof. Take $X = \mathsf{Path}_\infty(N)$, f the left shift, R the first-coordinate readout, and $L(y)$ the set of legal paths whose first coordinate is y. Each $L(y)$ is nonempty because every state has a successor. The readout condition is immediate. Every shifted path starting at y begins at a successor $z \in N(y)$, while any legal path starting at such z can be preceded by y. Thus $f[L(y)] = \bigcup_{z \in N(y)} L(z)$. □

The shift-space construction is classical. The new purpose here is to place the set-valued and probability-valued interfaces side by side without conflating them.

4 A finite hidden-state characterization and minimum cost

Hereafter Y is finite and K is any stochastic matrix. Let $G_K = (Y, A)$ denote its positive-transition digraph, with $(y, z) \in A$ iff $K(y, z) > 0$. An *infinite legal path* is a sequence $w = (y_0, y_1, \ldots)$ following edges of G_K. Write $\mathsf{Path}_\infty(K)$ for the set of such paths, including their initial states, and $\mathsf{Word}_n(K)$ for the length-n legal words. Define

$$B_n(K) = |\mathsf{Word}_n(K)|, \qquad m_*(K) = \inf\{|X| : (f, K, R, E) \text{ exact}, X \text{ finite}\},$$

with $\inf \varnothing = \infty$. A vertex is *cyclic* if it lies on a directed cycle, including a self-loop.

Lemma 4.1 (Finite-word lower bound)**.** *Every exact stochastic DNI interface with finite X satisfies $|X| \geq B_n(K)$ for every $n \geq 1$.*

Proof. By theorem 2.2, every legal length-n word has strictly positive probability as an observed prefix when preparation starts from its first state. Each hidden initial state x determines one and only one length-n output word. By (S1), preparatory supports for distinct initial observable states lie in disjoint readout fibres. Consequently, all $B_n(K)$ possible initial words require at least $B_n(K)$ distinct hidden initial states. □

Lemma 4.2 (Graph-path finiteness criterion)**.** *For a finite directed graph of nonzero outdegree at every vertex, the following are equivalent:*

(i) the set of all legal infinite paths is finite;

(ii) $\sup_n B_n < \infty$*;*

(iii) every vertex on a directed cycle has outdegree exactly one.

Proof. (i)⇒(ii): Each finite legal word extends to an infinite path, so each level has at most as many words as there are infinite paths. For (ii)⇒(iii), suppose a cyclic vertex v has two distinct successors and choose a directed cycle through v. For each integer $j \geq 0$, following that cycle exactly j times and then taking the other successor yields a legal prefix that can be extended forever. These infinitely many prefixes lead to pairwise distinct infinite paths, and for any prescribed q a sufficiently long common prefix length distinguishes the first q choices, contradicting bounded B_n. For (iii)⇒(i), every infinite path repeats a vertex in at most $|Y| + 1$ visited states. Its first repetition is on a directed cycle; all vertices of that cycle have unique successors, so the entire remaining infinite suffix is forced and periodic. There are only finitely many initial segments of length at most $|Y| + 1$, and therefore only finitely many infinite paths. □

Theorem 4.3 (Exact finite-resource characterization and minimality)**.** *For a finite-state stochastic kernel K on Y, the following conditions are equivalent:*

(a) K admits an exact stochastic DNI interface with finite deterministic hidden-state space X;

(b) every vertex on a directed cycle of G_K has exactly one positive-probability successor;

(c) $\sup_{n\geq 1} B_n(K)$ *is finite.*

When these conditions hold, the minimum hidden-state size exists and is

$$\boxed{m_*(K) = |\mathsf{Path}_\infty(K)| = \sup_{n\geq 1} B_n(K).} \tag{3}$$

If the conditions fail, $m_*(K) = \infty$ (meaning no finite *exact realization).*

Proof. If a finite exact interface exists, theorem 4.1 bounds all B_n by $|X|$. By theorem 4.2, this gives (b), (c), and finite $\mathsf{Path}_\infty(K)$. Conversely suppose that the graph condition holds. Take the *finite* hidden space $X = \mathsf{Path}_\infty(K)$, deterministic left shift f, first-coordinate readout R, and for each y let $E(y)$ be the Markov path law started at y, restricted to the finite set X. This is a probability measure because all possible paths eventually enter a deterministic cycle and hence belong to X. The path-space argument gives (S1)–(S2). Thus $m_*(K) \leq |X|$. For every pair of distinct infinite paths there is a finite length at which their prefixes differ. As X is finite, some common length n_0 separates all paths, and $B_{n_0} = |X|$. By theorem 4.1, every finite interface has size at least $|X|$. Hence equality holds, and monotonicity of the positive prefix counts yields the supremum formula. □

Corollary 4.4 (Minimum-cost possibilistic realization)**.** *For a finite relation $N : Y \to \mathcal{P}_+(Y)$, its minimum finite deterministic hidden-state size under (P1)–(P2) is $|\mathsf{Path}_\infty(N)|$ if every cyclic vertex has one successor, and no finite realization exists otherwise.*

Proof. The deterministic readout of a hidden state determines its infinite observation path. Conditions (P1)–(P2) imply by induction that every legal finite word is realized by a hidden state; hence $|X| \geq B_n(N)$ for every n. The graph lemma gives necessity. When legal infinite paths are finite, the construction of theorem 3.4 attains $|X| = |\mathsf{Path}_\infty(N)|$, while a sufficiently long word length distinguishes every pair of infinite paths and yields the matching lower bound. □

Remark 4.5 (Scope of the minimum). The optimality statement quantifies over freely chosen f, R, E and counts *states*, not bits of precision, preparation complexity, running time, physical plausibility, or cost of an independently prescribed source model. Enlarging the resource model changes the question.

5 A finite decision rule and worked examples

When the cycle condition holds, define $c(y)$ recursively as follows: on a cyclic vertex, set $c(y) = 1$; on each noncyclic vertex, set

$$c(y) = \sum_{z:K(y,z)>0} c(z). \tag{4}$$

The induced graph on noncyclic vertices is acyclic, so this recurrence can be evaluated in reverse topological order. Then

$$m_*(K) = \sum_{y\in Y} c(y). \tag{5}$$

Here $c(y)$ counts infinite legal paths beginning at y. Testing strongly connected components and directed self-loops decides the cycle condition; hence the minimum is computable without enumerating all hidden machines.

Example 5.1 (An optimal four-state interface)**.** Let $Y = \{a, b, c\}$ with

$$K(a, b) = \tfrac{1}{3}, \qquad K(a, c) = \tfrac{2}{3}, \qquad K(b, b) = K(c, c) = 1,$$

and all other transition probabilities zero. There are precisely four infinite legal paths:

$$w_1 = abbb\cdots, \quad w_2 = accc\cdots, \quad w_3 = bbb\cdots, \quad w_4 = ccc\cdots.$$

Set $X = \{w_1, w_2, w_3, w_4\}$, let f shift each path ($f(w_1) = w_3$, $f(w_2) = w_4$, $f(w_3) = w_3$, $f(w_4) = w_4$), and let R read its initial symbol. Take

$$E(a) = \tfrac{1}{3}\delta_{w_1} + \tfrac{2}{3}\delta_{w_2}, \quad E(b) = \delta_{w_3}, \quad E(c) = \delta_{w_4}.$$

Then $f_*E(a) = \frac{1}{3}E(b) + \frac{2}{3}E(c)$ and (S1)–(S2) hold. Since $c(a) = 2$ and $c(b) = c(c) = 1$, theorem 4.3 proves that *four* hidden states are necessary and sufficient.

Example 5.2 (Almost-sure absorption does not imply finite realization)**.** Take $Y = \{a, b\}$ with $K(a, a) = K(a, b) = 1/2$ and $K(b, b) = 1$. The process starting at a is eventually absorbed at b with probability one. Nevertheless, a lies on a self-loop and has two successors. Every integer $j \geq 0$ yields a different legal path $a^{j+1}b^\infty$ of positive probability. By theorem 4.3, there is no finite exact deterministic interface, despite almost-sure absorption.

Example 5.3 (Endlessly stochastic two-state flips)**.** For $Y = \{0, 1\}$ and $K = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{smallmatrix}\right)$, both vertices lie on cycles and have two successors. No finite exact deterministic interface exists. There is, however, an infinite one: take $X = \{0, 1\} \times [0, 1]^{\mathbb{N}_0}$, let R read the bit, let $E(s)$ prepare s with an independent uniform sequence, and define $f(s, u_0, u_1, \ldots) = (s \oplus \mathbf{1}_{\{u_0<1/4\}}, u_1, u_2, \ldots)$. Independence of the tail verifies (S1)–(S2).

6 Feasibility for a fixed hidden implementation

Proposition 6.1 (Linear feasibility with fixed dynamics)**.** *Suppose X, Y are finite and $f : X \to X$, $R : X \to Y$, and K are prescribed. An exact stochastic DNI interface exists if and only if real variables e_{yx} satisfy*

$$\begin{aligned} & e_{yx} \geq 0, && \sum_{x \in X} e_{yx} = 1, \\ & e_{yx} = 0 \quad \textit{when } R(x) \neq y, \\ & \sum_{u : f(u) = x} e_{yu} = \sum_{z \in Y} K(y, z) e_{zx} && \textit{for all } y \in Y, x \in X. \end{aligned}$$

If all $K(y, z)$ are rational, feasibility admits a rational solution E.

Proof. Set $e_{yx} = E(y, \{x\})$. The first three constraints are exactly stochastic normalization and (S1), and the final constraints are the x-coordinates of (S2). Conversely these equations define the required kernel. With rational K, the constraints define a bounded rational polytope; every nonempty bounded rational polytope has a rational point. □

This result distinguishes *freely chosen* finite realizations (governed by theorem 4.3) from the feasibility of a fixed deterministic implementation. It does not solve the general minimization problem when additional semantic restrictions are imposed.

7 Approximate stochastic interfaces

Let $\|\mu - \nu\|_{\mathrm{TV}} = \sup_A |\mu(A) - \nu(A)|$. Suppose (S1) holds exactly and for some $\varepsilon \geq 0$,

$$\sup_{y \in Y} \|f_* E(y) - (EK)(y)\|_{\mathrm{TV}} \leq \varepsilon. \tag{6}$$

Theorem 7.1 (Uniform marginal error bound)**.** *Under these assumptions, for each $n \in \mathbb{N}_0$,*

$$\sup_y \|R_* f_*^n E(y) - K^n(y)\|_{\mathrm{TV}} \leq \min\{1, n\varepsilon\}.$$

Proof. Put $\Delta_n = \sup_y \|f_*^n E(y) - EK^n(y)\|_{\mathrm{TV}}$. We have $\Delta_0 = 0$; by adding and subtracting $f_*^n EK(y)$, and using contraction of total variation under deterministic maps and Markov kernels, $\Delta_{n+1} \leq \varepsilon + \Delta_n$. Hence $\Delta_n \leq n\varepsilon$. Readout contracts total variation and (S1) gives $R_* EK^n = K^n$. The bound by one is automatic. □

This assertion concerns single-time *marginal* laws. It does not claim an equality or estimate of entire path distributions in the approximate case.

8 What a theory of theories still requires

A higher-level interface must specify signatures Σ_D, Σ_P, a shared signature Σ_0, translations $\Sigma_0 \to \Sigma_D$ and $\Sigma_0 \to \Sigma_P$, model classes, and their satisfaction relations. In a suitable category of signatures the pushout

$$\Sigma_D \sqcup_{\Sigma_0} \Sigma_P$$

is a *syntactic* object; it is not automatically a consistent joint semantic model nor a lowest-cost resource interface. Institution theory gives an established formal setting for such comparisons [5].

Proposition 8.1 (Elementary joint-model criterion for disjoint extensions)**.** *Suppose Σ_D and Σ_P are many-sorted first-order signatures whose intersection is precisely Σ_0, with no cross-signature axioms beyond those in their separate theories. Then their union theory has a model if and only if there exist a Σ_D-model M_D and a Σ_P-model M_P whose Σ_0-reducts are isomorphic.*

Proof. A joint model restricts to two models with literally equal shared reducts. Conversely, transport one model along the specified isomorphism of its shared reduct so that the shared domains and common symbols agree; interpret the nonshared symbols independently from the two models on the resulting common carriers. Because the nonshared signatures are disjoint and each theory's axioms use only its own signature, the combined structure satisfies both theories. □

Remark 8.2. Theorem 8.1 is an elementary amalgamation observation, *not* a guarantee that two independently prescribed dynamical systems admit any specific (S1)–(S2) interface. Those equations are additional mixed constraints, and semantic compatibility under them is strictly more demanding.

An unrestricted physical or operational theory may require contextual observables, richer state/effect formalisms, nonclassical composition, or resource measures other than $|X|$. Markov categories organize deterministic and stochastic morphisms [3]; generalized probabilistic theories provide another established operational language [4]. Neither generality is derived from the finite classical results above.

Open problem (restricted, precise). Given two independently fixed dynamical models, a shared observation signature, and a declared class of admissible translations and auxiliary states, determine whether an exact compatibility witness exists and, if it does, minimize an explicitly stated resource cost. Our finite freely-chosen-source special case is solved by theorem 4.3; this general fixed-source problem is not.

9 Interpretation, audit boundaries, and conclusions

The article makes a precise distinction between (1) translating deterministic transitions into stochastic or possibilistic transitions, (2) constructing realizations after choosing an unrestricted or resource-constrained hidden state, and (3) reconciling *independently specified* theories under fixed semantic obligations. Results about (1) and (2) do not establish (3). The finite-resource characterization is a concrete theorem about (2), and the fixed-implementation linear test is a tractable subproblem of (3).

Priority statement. The Dirac inclusion, path-space shift realization, Rogers–Pitman intertwining, and the broad categorical frameworks are established antecedents. The finite-resource characterization and recurrence are supplied here with self-contained proofs as a focused contribution within this working note; no claim of novelty relative to the entire literature follows without a broader priority review.

Philosophical limitation. No statement above resolves whether probabilities are ontological or epistemic, whether deterministic physical models exist for specific stochastic theories, or whether determinism entails any claim about free will or ethics. No special physical theory has been introduced.

Claim	Proven status / dependency
(S1)–(S2), observable Markov property	Established intertwining specialization; proved and attributed here.
Set-valued interface and support bridge	Finite-state formalization; propositions proved here.
Finite hidden-space existence and minimum	Exact necessary/sufficient graph criterion, formula, constructive proof.
Fixed hidden implementation	Explicit linear feasibility formulation.
General semantic theory amalgamation	Elementary disjoint-signature criterion only; broad goal open.
Physics, free will, ethical conclusions	None claimed.

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Zip–Inverse-Zip Interface Theory

Structural Recovery, Certified Transport, and Compositional Interfaces

Research exposition

20 October 2026

Abstract

Provenance-preserving zipping combines mathematical sources without conflating their identities. Inverse zipping recovers source records, whereas certified interfaces transport mathematical judgments. We establish elementary recovery and composition results, each with a formal corollary, and distinguish these structural facts from the substantive verification requirements of the Foggy Multiarticle Metaworld and OP-01.

1 Introduction

The Foggy Multiarticle Metaworld consists of mathematical source families whose definitions, hypotheses, theorems, and proof records must remain distinguishable even when studied jointly. Multiarticle zipping combines sources without identifying mathematical objects prematurely. Inverse zipping recovers their original contributions. Deterministic–Nondeterministic Interface Theory (DNI) raises a further question: under what conditions can results be transported between constituent theories?

$$\boxed{\text{Structural inclusion} \neq \text{certified theorem transport.}}$$

A source may be faithfully embedded in a combined research record without its theorems becoming theorems of another source.

2 Provenance-Preserving Zipping

Definition 2.1 (Mathematical island)**.** *A mathematical island is a structured record $T_i = (\Gamma_i, F_i, P_i, V_i, \nu_i)$, where the components represent hypotheses, claims, proof records, verification obligations, and provenance. Let A_i be its set of records.*

Definition 2.2 (Provenance-preserving zip)**.** *For $T_1, \dots, T_n$, define the structural zip as the tagged disjoint union*

$$Z = \coprod_{i=1}^{n} A_i = \{(i, a) : a \in A_i\}.$$

Thus $(i, a) = (j, b)$ precisely when $i = j$ and $a = b$.

Theorem 2.3 (Structural Recovery)**.** *For every i there are an injective inclusion $\iota_i : A_i \hookrightarrow Z$ and an inverse-zip map $\pi_i : \iota_i(A_i) \to A_i$ satisfying*

$$\boxed{\pi_i \circ \iota_i = \mathrm{id}_{A_i}.}$$

Proof. Set $\iota_i(a) = (i, a)$ and $\pi_i(i, a) = a$. Uniqueness of tags makes the inclusion injective and the projection well-defined on its stated domain. Therefore $\pi_i(\iota_i(a)) = a$. □

Corollary 2.4 (Lossless Source Recovery)**.** *Every original record is recovered without alteration: $\pi_i(\iota_i(a)) = a$ for all $a \in A_i$. Distinct tagged source records cannot be identified by the zip.*

A zip is a shared workspace, not an automatic merger of theories. Its projections need not act on records originating in other islands.

3 Certified Mathematical Interfaces

Definition 3.1 (Proof-preserving interface)**.** *Let T_i, T_j have derivability relations $\vdash_i, \vdash_j$. A proof-preserving interface $I_{ij} : T_i \to T_j$ comprises a well-typed translation of formulas and hypotheses and a computable proof-certificate transformation such that*

$$\Gamma \vdash_i \varphi \quad \Longrightarrow \quad I_{ij}(\Gamma) \vdash_j I_{ij}(\varphi),$$

with the transformed certificate accepted by the designated destination verifier.

Theorem 3.2 (Certified Transport)**.** *If I_{ij} is a proof-preserving interface and p proves $\Gamma \vdash_i \varphi$, then $I_{ij}(p)$ proves $I_{ij}(\Gamma) \vdash_j I_{ij}(\varphi)$.*

Proof. Apply the certificate transformation stipulated in the definition. Its output establishes the translated judgment and is accepted by the destination verifier. □

Corollary 3.3 (Conditional Theorem Preservation)**.** *If $T_i \vdash \varphi$ and the translated hypotheses are discharged or established in T_j, then*

$$\boxed{T_j \vdash I_{ij}(\varphi).}$$

Zipping provides a location for comparison; the certified bridge justifies transport. In general $\pi_j \circ \iota_i$ is undefined when $i \neq j$.

4 Composition of Interfaces

Definition 4.1 (Composable interfaces)**.** *Proof-preserving interfaces $I_{12} : T_1 \to T_2$ and $I_{23} : T_2 \to T_3$ are composable when the first interface's output judgments and certificates meet the second's input, hypothesis, and verification requirements.*

Theorem 4.2 (Interface Composition)**.** *For composable proof-preserving interfaces, $I_{13} = I_{23} \circ I_{12}$ is proof-preserving.*

Proof. Given $\Gamma \vdash_1 \varphi$, apply I_{12} to obtain $I_{12}(\Gamma) \vdash_2 I_{12}(\varphi)$, then apply I_{23} to obtain

$$I_{23}(I_{12}(\Gamma)) \vdash_3 I_{23}(I_{12}(\varphi)).$$

Composing the computable certificate transformations yields a verified destination certificate under the composability assumptions. □

Corollary 4.3 (Certified Path Transport)**.** *For any finite directed path of composable proof-preserving interfaces*

$$T_1 \xrightarrow{I_{12}} T_2 \xrightarrow{I_{23}} \cdots \xrightarrow{I_{(n-1)n}} T_n,$$

the composite $I_{1n} = I_{(n-1)n} \circ \cdots \circ I_{12}$ preserves derivability.

A network can support indirect theorem transport without a direct interface between every pair of islands. A structural zip-tree edge, however, is not automatically proof-preserving.

5 Recursive Zipping and Interface Networks

For example, let $Z_{12} = T_1 \bowtie T_2$, $Z_{34} = T_3 \bowtie T_4$, and $Z_{1234} = Z_{12} \bowtie Z_{34}$. Each leaf has a tagged inclusion into the final record and a recovery projection on its image. Structural edges preserve records; certified interface edges preserve specified mathematical judgments.

Theorem 5.1 (Recursive Recovery)**.** *If a finite recursive sequence of provenance-preserving zips retains all original tags and records, every original island has maps ι_i into the final zip and π_i on its image such that*

$$\boxed{\pi_i \circ \iota_i = \mathrm{id}_{A_i}\,.}$$

Proof. Induct on the zip tree. At each leaf recovery is the identity. At each internal node the tagged-union inclusion and projection satisfy Structural Recovery. Compose these maps along the unique leaf-to-root path. □

Corollary 5.2 (Order-Independent Source Recovery)**.** *Two recursive zip trees on the same tagged source records yield canonically bijective final source-record unions, by the identity on tags and records. The order of zipping does not affect recoverability, although it may affect search schedules, intermediate deductions, and costs.*

6 DNI Obstructions

DNI studies when interfaces cannot exist under specified preservation and resource requirements. In particular, its finite-state exact-realization obstructions motivate the distinction

$$\boxed{\text{Structural compatibility} \not\Rightarrow \text{exact dynamical realizability.}}$$

Two islands may zip successfully while lacking an exact finite-resource realization. An obstruction to exact behavior preservation does not, without further argument, exclude weaker reachability- or certificate-preserving interfaces.

7 Application to OP-01

The original OP-01 requires a verifier-gated, fully costed settlement making material use of at least 26 families from its frozen 29-family corpus. DNI's later registration as N38 does not alter that requirement.

Let $\mathcal{G} = (V, E)$ be a directed graph whose vertices are original source families and whose edges are certified interfaces. An admissible expedition would need to establish:

1. Material contribution from at least 26 original families.
2. Certified translation and proof transport across every utilized cross-island edge.
3. A valid verifier-accepted settlement.
4. Full accounting of search, translation, activation, verification, and bookkeeping costs.
5. Improvement over the declared fair comparator.

A tree on 26 vertices has 25 edges, but undirected connectivity alone guarantees neither the necessary transport directions nor material proof dependencies. The resulting optimization may be a directed dependency-hypergraph problem rather than ordinary shortest-path search.

8 Conclusion

$$\begin{aligned}
\text{Zip} &\longrightarrow \text{provenance-preserving combination},\\
\text{Inverse zip} &\longrightarrow \text{exact source recovery},\\
\text{Certified interface} &\longrightarrow \text{valid theorem transport},\\
\text{Composition} &\longrightarrow \text{multi-island transport},\\
\text{DNI obstruction} &\longrightarrow \text{limits of realizability}.
\end{aligned}$$

The recovery and composition results are elementary consequences of their explicit definitions, not claims of independently machine-verified discoveries. The substantive next question is when a provenance-preserving zip admits a certified cross-island interface, rather than merely recovering its original components.

ATLAS Repair and Re-Audit

Deterministic–Nondeterministic Interfaces, Version 0.2

Proof, mathematical scope, priority, and typesetting review • 20 October 2026

Verdict: PASS for the repaired, restricted finite-classical mathematics.
Still preliminary: This is *not* an established general interface theory of all independently specified deterministic and probabilistic theories. The stronger finite-resource result is fully proved in the repaired text, but an exhaustive literature priority audit has not been completed.

1. Materials and audit standard

Compared the original eight-page article *DNI v0.1* and prior *ATLAS Audit v0.1* with the repaired eight-page source and rendered PDF, *DNI v0.2 ATLAS Repaired*. Reconstructed the proofs independently, checked the finite-support and state-count arguments, performed exhaustive three- and four-state graph checks and exact rational witness verification, validated cited precedent for the intertwining identities, and inspected all rendered PDF pages. This is an ATLAS-style mathematical audit conducted in this session, not a claim of execution by any separately running research system.

2. Disposition of all major earlier findings

ID	Status	Repair and evidence
M1: Priority	RESOLVED	(S1)–(S2) explicitly identified with $\Lambda\Phi = I$, $\Lambda P = K\Lambda$; Rogers–Pitman (1981) and Alberts cited before the Markov-history theorem. No originality claim on those identities.
M2: Possibility	RESOLVED	Added (P1)–(P2), finite-support bridge, support-composition proposition, shift-path existence, and a finite-state minimum corollary. The distinction between nondeterminism and stochastic weights is maintained.
M3: Minimality	RESOLVED (restricted)	Defined $m_*(K)$ over finite hidden spaces and proved exact necessary and sufficient graph criterion and minimal cardinality. General categorical/semantic minimality deliberately remains open.
M4: Unlimited encoding	RESOLVED (restricted)	Added finite-resource obstruction, minimum hidden-state recurrence, fixed-implementation LP, and two contrasting examples. The freedom to choose f, R, E is explicit.

3. Independent mathematical verification

Repaired result	Result	Independent check
Stochastic DNI and Markov histories	PASS	E concentrated on readout fibres and $f_*E = EK$ suffice for conditional hidden distributions to renew at each observed step, giving the full Markov joint law. Established Rogers–Pitman specialization.

Repaired result	Result	Independent check
Finite support bridge	PASS	For finite nonnegative sums, positive support of $EK(y)$ equals the union of supports of $E(z)$ for all positive $K(y,z)$. Pushforward support under a deterministic map equals its image.
Compositional support	PASS	Positivity of a finite matrix product is equivalent to the existence of a positive summand.
Possibilistic path shift	PASS	Every legal tail $z, z_1, \ldots$ may be prepended by an allowed predecessor y; therefore $f[L(y)] = \bigcup_{z \in N(y)} L(z)$.
Positive-word lower bound	PASS	A fixed finite hidden initial state fixes one output word; different initial readouts occupy disjoint preparation fibres. Every positive Markov word must occur. Hence $\lvert X\rvert \geq B_n$.
Graph-path equivalence	PASS	Cyclic branching permits arbitrarily delayed exits and yields unbounded distinct prefixes; without cyclic branching every walk is forced after its first repeated vertex, so only finitely many infinite paths exist.
Exact minimal realization	PASS	Path-space shift gives a witness when finite; the word bound is tight at a length separating all finitely many distinct infinite paths. Thus $m_* = \lvert\mathsf{Path}_\infty\rvert = \sup_n B_n$.
Possibilistic minimum	PASS	(P2) lifts every admissible finite word inductively to a hidden starting state, giving the same lower bound; finite shift-path realization attains it.
Counting recurrence	PASS	Every cyclic vertex has one forced infinite continuation; every noncyclic vertex splits legal paths by first successor. Noncyclic induced subgraph is acyclic.
Four-state witness	PASS	All 12 intertwining coordinates and nine readout coordinates were independently checked using exact rational arithmetic. Four paths give the matching lower bound.
Fixed f, R LP	PASS	Normalization, readout fibres and the coordinate form of $f_*E = EK$ are linear in E; rational coefficients admit a rational feasible point when nonempty.
Finite-time TV stability	PASS	Reconstructed telescoping/one-step contraction bound $\Delta_{n+1} \leq \Delta_n + \varepsilon$; it is marginal-only as written.
Elementary joint-model criterion	PASS	In a disjoint extension of a common signature, models can be transported across an isomorphism of common reducts and glued without mixed axioms. No stronger amalgamation theorem claimed.

4. Independent computational checks (not proof substitutes)

1. **Exhaustive three-state support graphs:** 343 directed graphs with all three outdegrees nonzero were enumerated ($7^3 = 343$). Exactly 39 satisfied the no-branching-on-cycles condition and 304 failed. In every satisfying case, length-11 path-word counts had stabilized and matched the graph recurrence. Every failing case had strictly more positive length-11 words than length-2 words. This checks many instances but is not an exhaustive verification of arbitrary state counts.

2. **Independent four-state exhaustive check:** All $15^4 = 50{,}625$ four-vertex directed graphs with nonempty outdegree were separately enumerated. Cycle membership was computed using positive-length transitive closure rather than the depth-first check above. Exactly 904 satisfied the graph criterion and 49,721 failed. Each satisfying graph had stabilized path-word counts by length 12; each failing graph

had more length-12 words than length-4 words. This provides a second finite consistency check, not a proof of the general theorem.

3. **Four-hidden-state example:** The rational interface with $E(a) = \frac{1}{3}\delta_{w_1} + \frac{2}{3}\delta_{w_2}$, $E(b) = \delta_{w_3}$, $E(c) = \delta_{w_4}$ was checked entry-by-entry for (S1) and (S2).

4. **Typesetting:** The revised TeX compiles to eight letter-size PDF pages with no overfull boxes, LaTeX errors, or undefined references in the final pass. All eight pages were rendered and reviewed; two minor underfull paragraph diagnostics do not conceal mathematics.

5. Remaining qualifications and publication advice

Q1 — Originality: NOT CERTIFIED. The newly organized finite-state characterization is proved, but elementary path/cycle classification and automata realizations may have earlier formulations. The audit establishes mathematical correctness, not international literature priority. Present the result as a self-contained finite-classical theorem pending extended literature review.

Q2 — Theory-of-theories scope: LIMITED. The paper formally states a joint-model criterion for disjoint first-order signatures; it does not construct a universal mixed-signature interface or prove conservative embedding of arbitrary input theories. A syntactic pushout need not yield semantic compatibility.

Q3 — Theoretical costs: QUALIFIED. m_* optimizes the number of hidden states over freely chosen deterministic f, readout R, and preparation E. It does not minimize bits of storage, probability-description complexity, physical implementability, runtime, or a preassigned source deterministic theory.

Q4 — Semantics: LIMITED. Probability is classical and discrete time; no nonclassical measurement, contextuality, causal locality or physical reconstruction theorem is established. The role of statistical preparation remains declared, rather than suppressed.

Q5 — Algorithmic precision: ACCEPTABLE. The exact recurrence is finite and testable. Bit complexity and optimization when the source structures are independently fixed remain future work.

6. Final certification statement

Mathematical proofs: PASS for results within stated hypotheses.
Prior ATLAS M1–M4 repairs: PASS in the expressly restricted sense above.
Priority and broader philosophical/physical applicability: NOT CERTIFIED.
Document quality: PASS.
Recommended designation: *ATLAS-audited working mathematical article, Version 0.2.* The parent Version 0.1 remains unchanged; no publication action was taken.

References checked

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2. T. Alberts, "Pitman–Rogers on Finite State Markov Chains," University of Utah, https://www.math.utah.edu/~alberts/notes/sle-markov-functions/discrete-pitman-rogers/.

3. T. Fritz, "A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics," *Advances in Mathematics* **370**, 107239 (2020).

4. J. A. Goguen and R. M. Burstall, "Institutions: Abstract model theory for specification and programming," *Journal of the ACM* **39**(1), 95–146 (1992).